

MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§4.1

Divisibility

Shortcuts to decide whether one number is divisible by another

Today's Goal

After completing this section, you should be able to:

- Explain what it means for an integer to be divisible by a nonzero whole number, and connect divisibility to factors and to division without remainder
- Apply the divisibility tests for 2, 3, 4, 5, 6, 8, 9, 10, and 11 to decide whether a number is divisible without carrying out the division
- Apply the three divisibility tests for 7—the Standard Test, Chika's Test, and Mo's Test—repeatedly until the number that remains is small enough to judge
- Use a divisibility test in reverse to find a missing digit that makes a number divisible by a given divisor

Learning Target 10

LT.10 Divisibility

I understand what it means for an integer to be divisible by another integer and can apply the standard divisibility tests for 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11.

What Is Different?

Consider the expressions $15 \div 5$ and $15 \div 4$. What is different about them?

There are three groups of 5 in 15. So $15 \div 5 = 3$, and we can say that “5 divides evenly into 15.”

There are three groups of 4 in 15, but there will be a remainder of 3. So “4 does not divide evenly into 15.”

Divisible

Definition 4.1.2

An integer n is **divisible** by a nonzero whole number d if d divides evenly (i.e., without remainder) into n . In other words, n is divisible by d if d is a factor of n . Equivalently, n is divisible by d if $n = d \cdot k$ for some integer k .

Divisibility Tests for 2 and 5

Theorem 4.1.4 Divisibility Test for 2

An integer n is divisible by 2 if and only if the units digit of n is 0, 2, 4, 6, or 8 (i.e., the units digit is even).

Theorem 4.1.10 Divisibility Test for 5

An integer n is divisible by 5 if and only if the units digit of n is either 0 or 5.

Divisibility Test for 10

Theorem 4.1.23 Divisibility Test for 10

An integer n is divisible by 10 if and only if the units digit is 0.

Divisibility Tests for 3 and 9

Theorem 4.1.5 Divisibility Test for 3

An integer n is divisible by 3 if and only if the sum of the digits of n is divisible by 3.

Theorem 4.1.21 Divisibility Test for 9

An integer n is divisible by 9 if and only if the sum of the digits of n is divisible by 9.

Divisible by 3? Divisible by 9?

Determine whether or not each number is divisible by 3 and by 9.

(a) 8,316

The sum of the digits is $8 + 3 + 1 + 6 = 18$. Since 18 is divisible by 3 and by 9, so is 8,316.

(b) 5,862

The sum of the digits is $5 + 8 + 6 + 2 = 21$. Since 21 is divisible by 3, so is 5,862. Since 21 is not divisible by 9, neither is 5,862.

Change One Digit

The number 527 is not divisible by 3. Change one of its digits to make a number that is divisible by 3. How many different ways can you do it?

The sum of the digits is $5 + 2 + 7 = 14$.

- Hundreds digit: 327, 627, 927 (3 ways)
- Tens digit: 507, 537, 567, 597 (4 ways)
- Units digit: 522, 525, 528 (3 ways)

There are $3 + 4 + 3 = 10$ ways.

Divisibility Test for 4

Theorem 4.1.8 Divisibility Test for 4

We have two divisibility tests for 4:

1. An integer n is divisible by 4 if and only if the number formed by the last two digits of n (i.e., the tens digit and the units digit) is divisible by 4.
2. An integer n is divisible by 4 if and only if the quotient of $n \div 2$ is even.

4,824: the last two digits form 24, which is divisible by 4. So 4,824 is divisible by 4.

Second test: $4,824 \div 2 = 2,412$, which is even.

Divisibility Test for 6

Theorem 4.1.11 Divisibility Test for 6

An integer n is divisible by 6 if and only if n is divisible by 2 and 3.

8,316 is even, and its digits add to 18. So 8,316 is divisible by 6.

12 is divisible by 2 and by 4. Is 12 divisible by 8?

No: $12 \div 8 = 1 \text{ R } 4$.

So being divisible by 2 and by 4 does not make a number divisible by 8. The test for 6 works because 2 and 3 share no factor.

Divisibility Test for 7

Theorem 4.1.13 Divisibility Test for 7

We have three divisibility tests for 7:

1. (Standard Test) Double the units digit of n and subtract the result from the number formed by the rest of the digits of n . n is divisible by 7 if and only if this difference is divisible by 7.
2. (Chika's Test) Multiply the units digit by 5 and add the result to the number formed by the rest of the digits of n . n is divisible by 7 if and only if this sum is divisible by 7.

Divisibility Test for 7, Continued

Theorem 4.1.13, continued

3. (Mo's Test) Triple the number formed by the last two digits of n and subtract the result from the number formed by the rest of the digits of n . n is divisible by 7 if and only if this difference is divisible by 7.

Is 8,316 Divisible by 7?

Determine whether or not 8,316 is divisible by 7.

Standard Test

$$831 - 2 \cdot 6 = 819$$

$$81 - 2 \cdot 9 = 63$$

Chika's Test

$$831 + 5 \cdot 6 = 861$$

$$86 + 5 \cdot 1 = 91$$

$$9 + 5 \cdot 1 = 14$$

Mo's Test

$$83 - 3 \cdot 16 = 35$$

63, 14 and 35 are all divisible by 7, so 8,316 is divisible by 7.

Divisibility Tests for 7 in MyOpenMath

Note 4.1.16

For questions involving divisibility tests for 7 in MyOpenMath, please use the Standard Test unless the directions say to do otherwise.

Test for 8 (for 3-Digit Numbers)

Theorem 4.1.17 Divisibility Test for 8 (for 3-digit numbers)

Let n be a 3-digit number.

1. Multiply the hundreds digit by 4.
2. Multiply the tens digit by 2.
3. Leave the ones digit alone.
4. Add all of the above.

n is divisible by 8 if and only if the sum from Step 4 is divisible by 8.

Divisibility Test for 8

Theorem 4.1.19 Divisibility Test for 8

An integer n is divisible by 8 if and only if the number formed by the last three digits of n is divisible by 8.

Determine whether or not 6,452 is divisible by 8.

We only need to check 452: $4 \cdot 4 + 2 \cdot 5 + 2 = 28$.

Since 28 is not divisible by 8, neither is 452, and neither is 6,452.

Divisibility Test for 11

Theorem 4.1.24 Divisibility Test for 11

We have two divisibility tests for 11:

1. Alternate between adding and subtracting the digits of n beginning by subtracting the second digit from the first digit (from left to right). n is divisible by 11 if and only if the result is divisible by 11.
2. Subtract the units digit from the number formed by the rest of the digits of n . n is divisible by 11 if and only if this difference is divisible by 11.

Is 8,316 Divisible by 11?

Determine whether or not 8,316 is divisible by 11.

First test: $8 - 3 + 1 - 6 = 0$, and 0 is divisible by 11.

Second test: $831 - 6 = 825$, then $82 - 5 = 77 = 7 \cdot 11$.

So 8,316 is divisible by 11.

Divisibility Tests

Which of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 is 5,148 divisible by?

Divisible by 2, 3, 4, 6, 9, and 11.

Not divisible by 5, 7, 8, or 10.